

the start and end dates, dependencies between tasks, and the critical path (the set of tasks that are dependent on each other, with delays in one of the tasks impacting the end date of the project). There are options for a project holiday calendar and per-resource holiday calendar, where the project manager can define holidays when no resource will work, and also the option of a separate calendar for a resource. For example, a software tester going on a two-week Himalayan trip during the last month of a project in test phase has a different impact than if his vacation was during the first half of the project.

Project planning and tracking basics

Proper planning is the first step towards successful project execution. Planning should be practical and detailed, and take into consideration all the constraints. The project manager plays a key role during planning, by asking the right questions on the effort it takes to complete a task, the dependencies with other members or teams, the potential delays involved and the risks that can be foreseen. Getting a buy-in from all project team members is the key to success. The project manager does not own the schedule—the team does.

Many projects use tools during the planning phase, but when it comes to tracking, these tools are abandoned. Tracking the project outside the tool poses problems, such as not knowing the impact of delayed tasks, or the tasks that are ahead of time. A tool that shows the critical path is handy here (see an example of the critical path in the next section). So, tracking using a tool like GanttProject is as essential as using it for planning.

One of the reasons many engineers don't have access to MS Project is due to the cost. The licence fee inhibits its use. A free and open source tool like GanttProject can be installed on every machine, and the Gantt file can be shared on a server using WebDAV.

Setting up GanttProject

GanttProject is available at <http://www.ganttproject.biz/> and also sourceforge.net. The latest version is 2.5.3. As with any other software installation, the best part of Windows is the wizard-based easy installation. Accept the licence agreement, click *Next* a few times, and then *Finish*; you're now ready to use it! The screen-shots in this article are of GanttProject 2.5.3 on Windows 7.

In this article, we will focus on creating a new project called the 'Sample GUI Project'. This is a software development project that uses the waterfall model, and goes through the typical requirements, design, coding and testing phases. We also have team members (resources) who will play different roles—the architects, developers, testers, etc.

A first look at GanttProject

The application has two panes: the left one lists tasks and their details; the right pane displays the schedule in Gantt format (or, when the user chooses the *PERT* option from the menu,

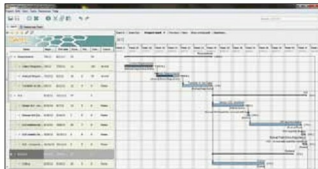


Figure 1: Application window with task and Gantt chart panes

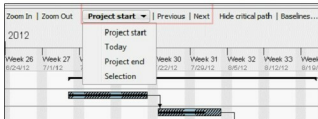


Figure 2: Control chart display using buttons above the chart

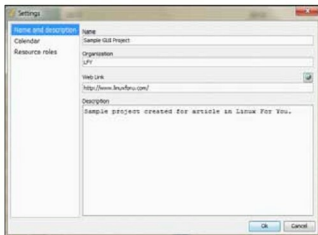


Figure 3: Project settings—entering project details

it projects the PERT chart). For more details on PERT charts, please refer to Wikipedia.

Like any other Windows application, you have a menu bar on top, a context-sensitive toolbar below it and a status bar. To vertically navigate the chart, scroll the mouse in the *Task pane* (left). Horizontal navigation of the chart is done by using the buttons and menu item just above the chart (Figure 2). Scrolling the mouse in the Gantt view alters the zoom. If there are a large number of tasks in the project, or the project is of a long duration, the Gantt chart will exceed the window size. To see the complete chart, one has to 'Zoom Out'. This can make the task bars in the chart too small to be legible, though. 'Zooming In' increases the granularity of the duration from months, to weeks and days.

Users can view portions of the project using the dropdown menu, which has options like *Project Start*, *Today*, *Project End* and *Selection*. *Project Start* will pan the Gantt